

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

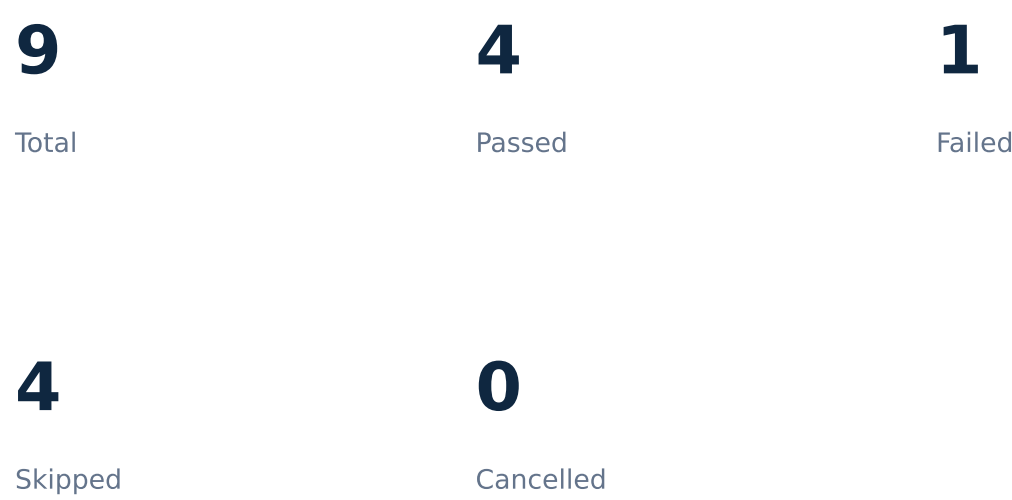
20260813_135732_e13866

Date

2026-08-13T12:57:32.167445+00:00

Executive Summary

20260813_135732_e13866



Overall measured availability: 66.07%

During this campaign, 4 of 9 scenarios passed, 1 failed and 4 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	N/A	1
memory	N/A	1
network-partition	N/A	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	2.719	12.672	True	True	10.992	16.818	11	15.38	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate failover routes require authentication, database prep	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	8.502	19.523	N/A	100.0	PASS	Database prep	N/A
memory	swarm	147.594	1.734	True	True	1023.342	250.903	1	99.21	PASS	Database prep	N/A
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database prep	N/A
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database prep	N/A
container-kill	kubernetes	6.672	18.094	True	True	1373.8	2061.282	7	22.22	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	N/A	N/A
cpu	kubernetes	N/A	N/A	N/A	N/A	13528.469	4008.091	4	93.55	FAIL	N/A	Stress pod finished with phase Running.

swarm — container-kill

State: **Pass**
Detection Time (s): 2.719

Recovery Time (s): 12.672

HTTP Success: N/A

HTTP Failures: 11

Availability (%): 15.38

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: **Pass**
Detection Time (s): 147.594

Recovery Time (s): 1.734

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 99.21

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

swarm — latency

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — container-kill

State: **Pass**
Detection Time (s): 6.672
~~Recovery Time (s): 18.094~~
HTTP Success: N/A
HTTP Failures: 7
Availability (%): 22.22
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

State: **Fail**
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 4

Availability (%): 93.55

CPU (%): N/A

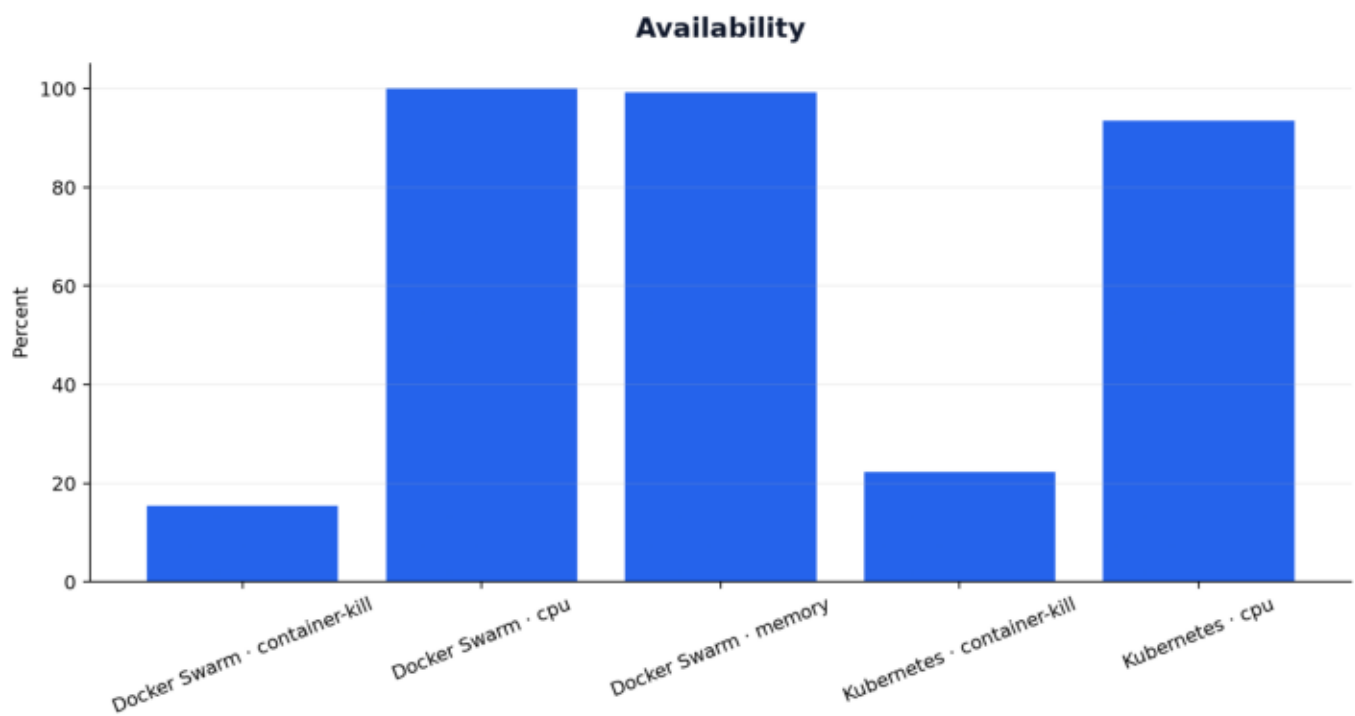
Memory (MB): N/A

Recorded interpretation

Stress pod finished with phase Running.

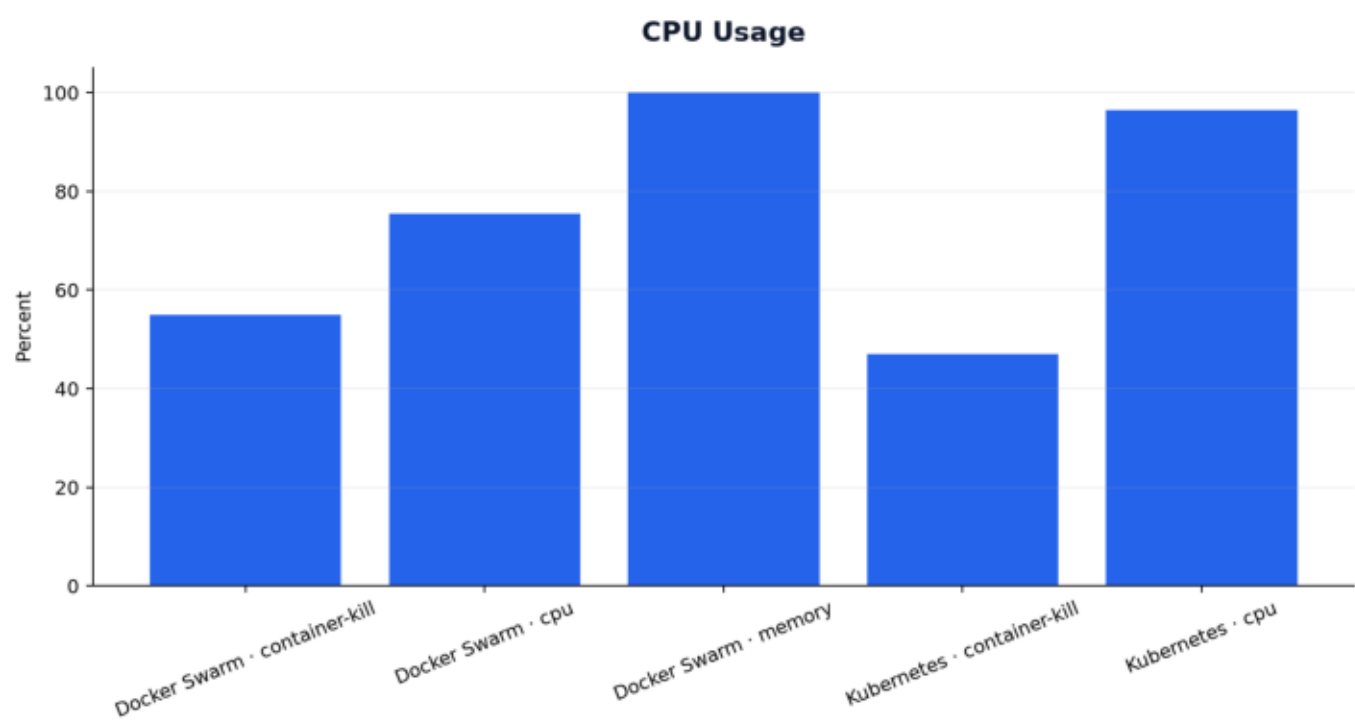
Availability

Current-session measured data



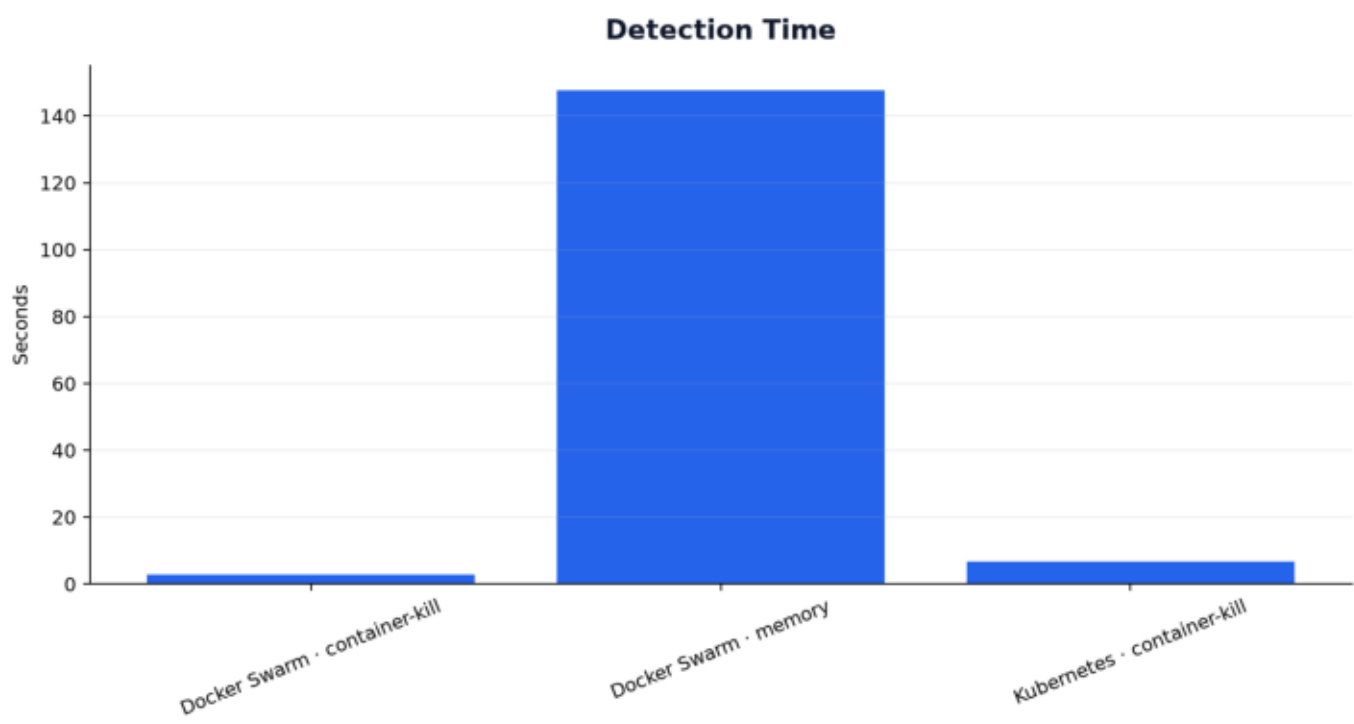
Cpu Usage

Current-session measured data



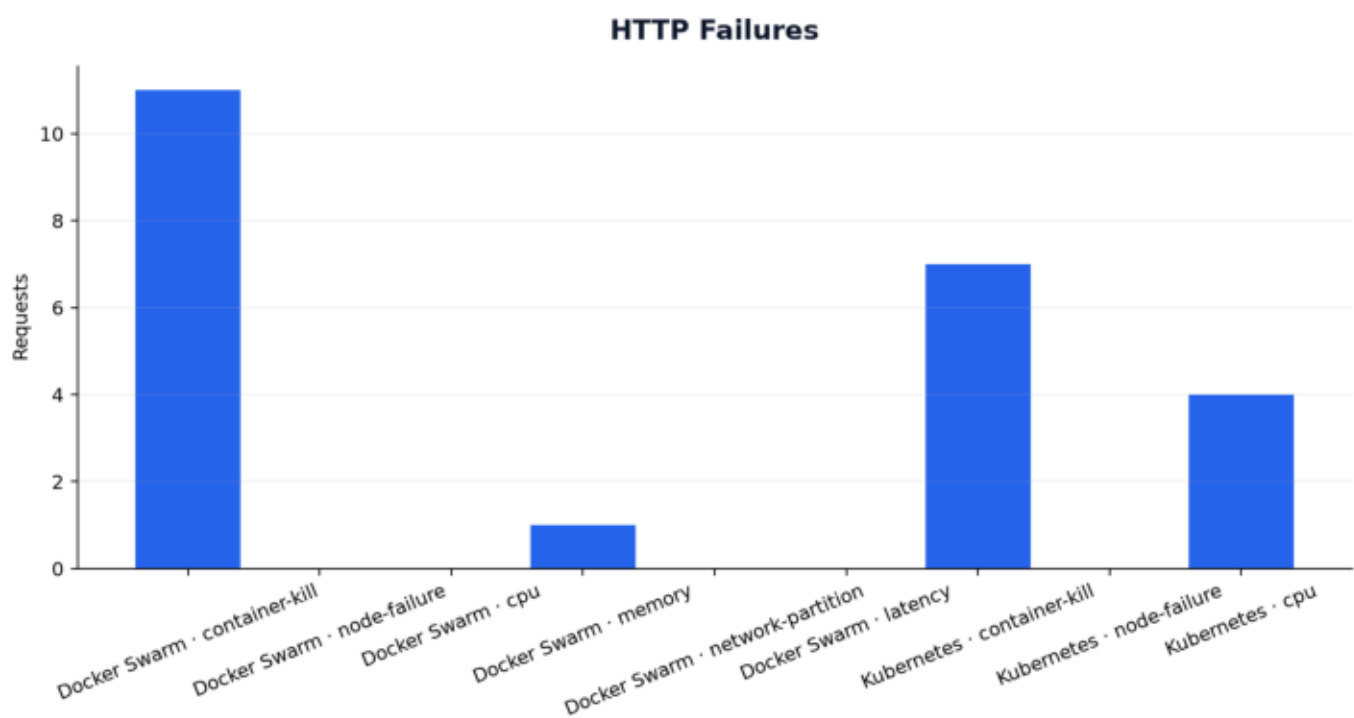
Detection Time

Current-session measured data



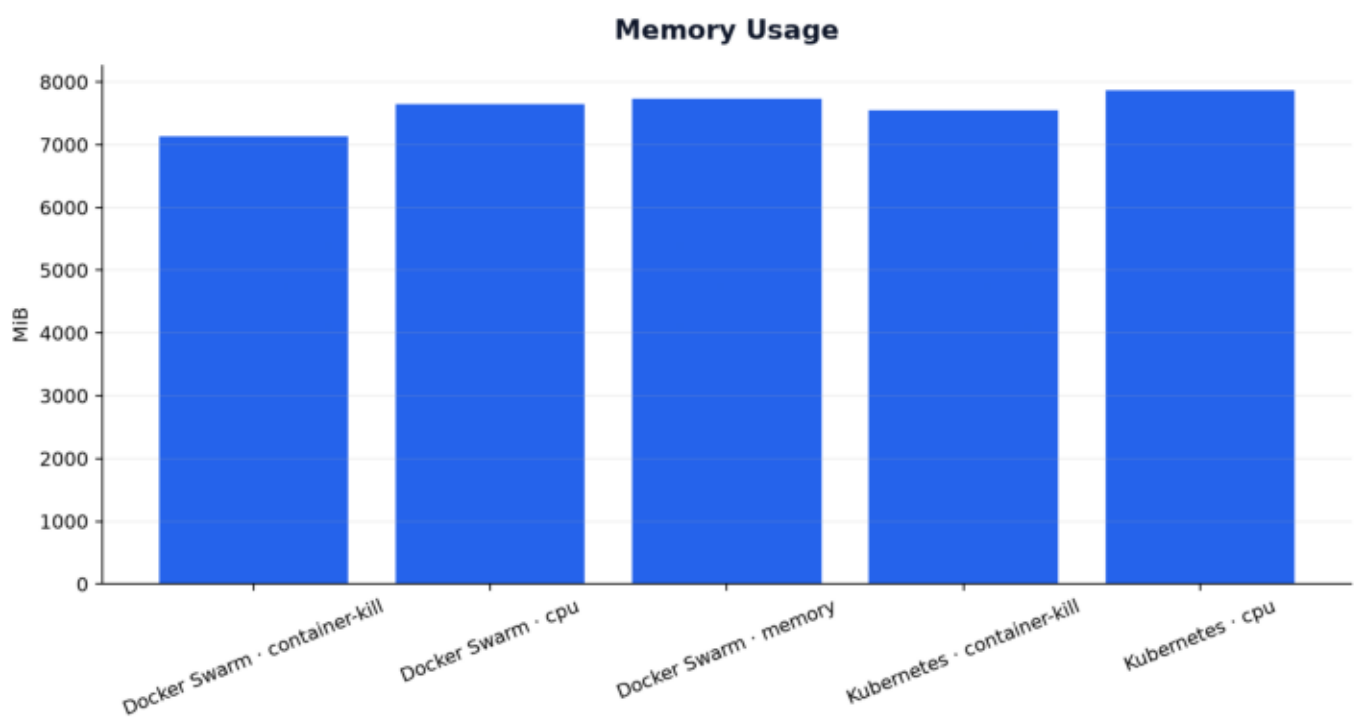
Http Failures

Current-session measured data



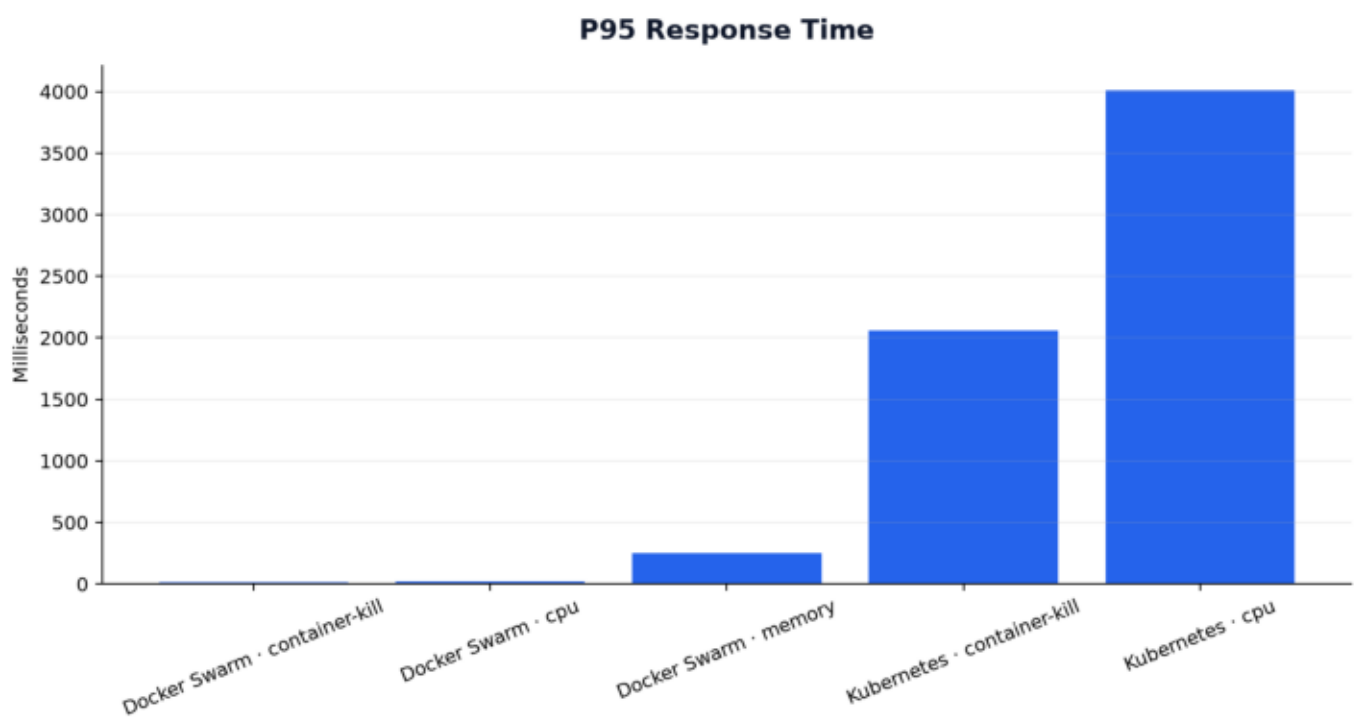
Memory Usage

Current-session measured data



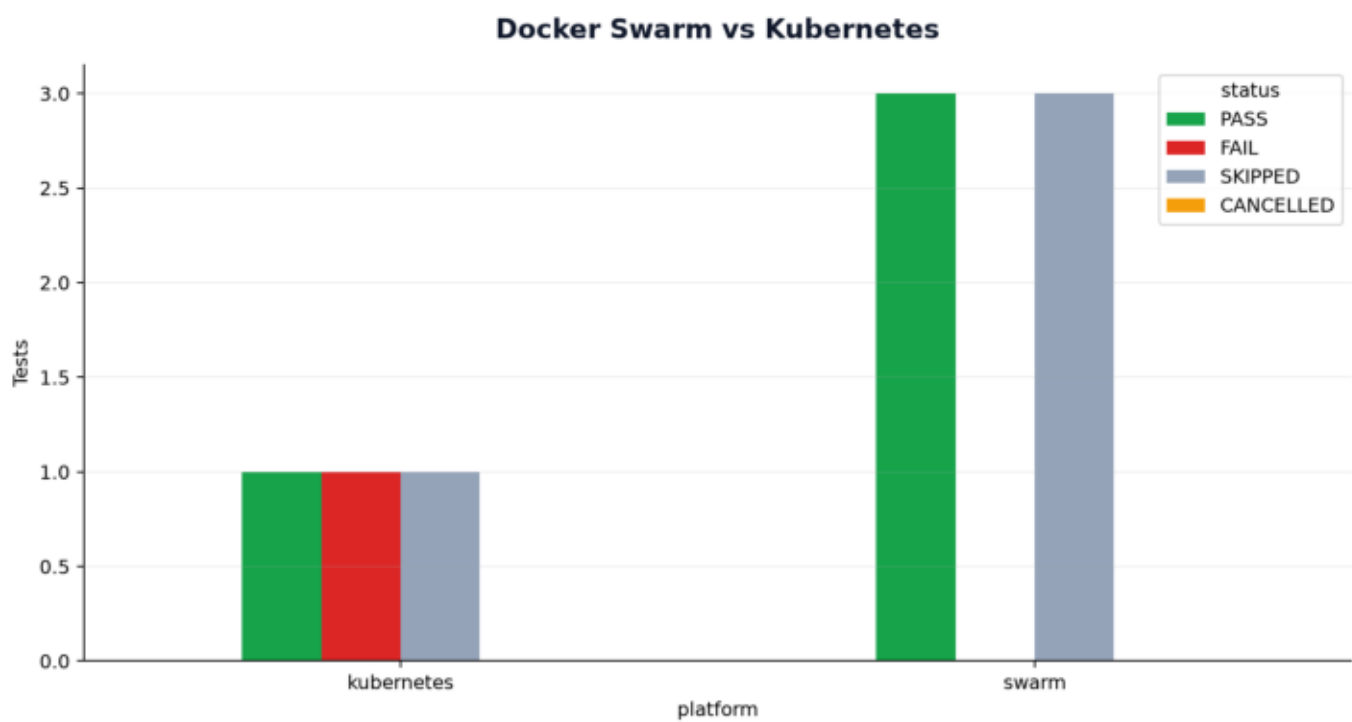
P95 Response Time

Current-session measured data



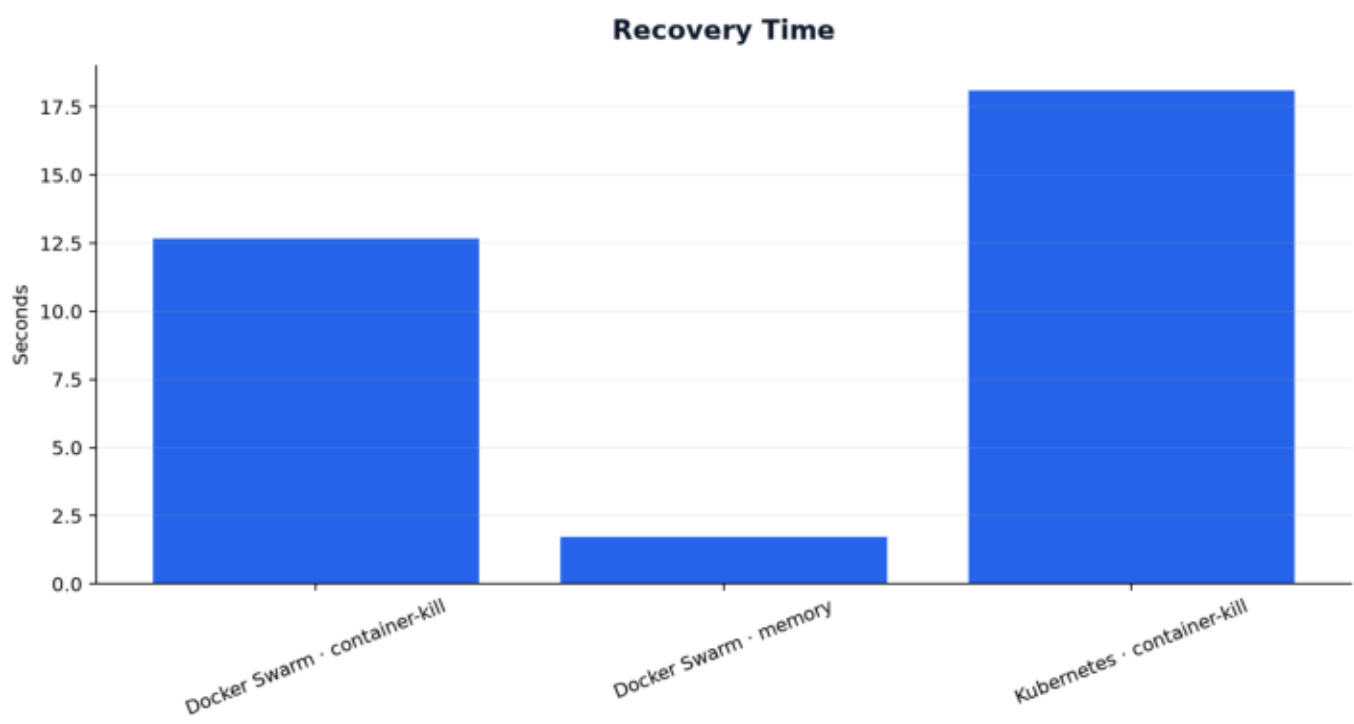
Platform Comparison

Current-session measured data



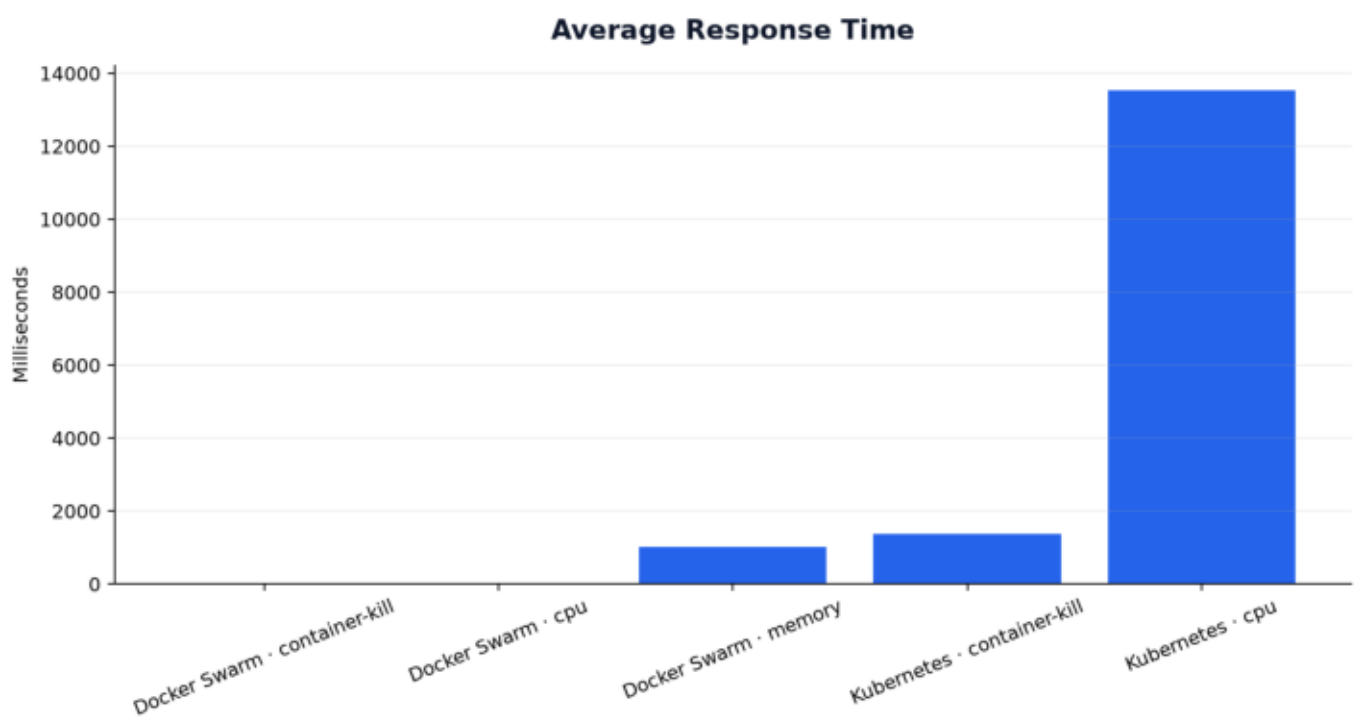
Recovery Time

Current-session measured data



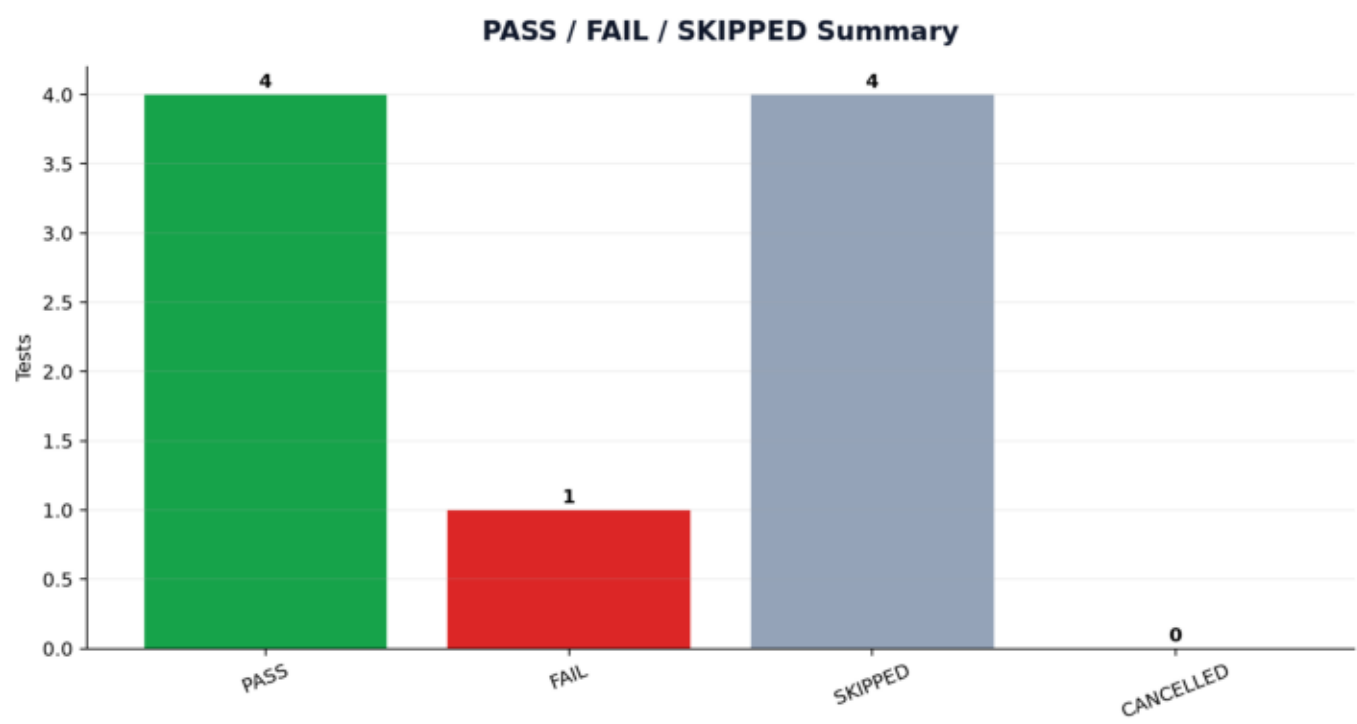
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
kubernetes	cpu	FAIL	Stress pod finished with phase Runn

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling. Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP
swarm	network-partition	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP
swarm	latency	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

Swarm vs Kubernetes Comparison

Platform	FAIL	PASS	SKIPPED
kubernetes	1	1	1
swarm	0	3	3

Final Conclusion

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During this campaign, 4 of 9 scenarios passed, 1 failed and 4 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.